Iterative image generation: from DRAW to denoising diffusion on MNIST

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Abstract

We study the progression from single-pass variational generation to iterative latent refinement to full denoising diffusion on MNIST. Starting from a vanilla VAE, we first show that replacing the single decoder pass with a recurrent DRAW canvas of $T = 10$ glimpses cuts the test negative ELBO from 102.9 to 93.8 nats per image at the same thirty-epoch budget. Adding Gaussian filterbank attention to DRAW requires twice the training time to converge on this small dataset, reaching 99.7 nats at sixty epochs. We then implement a Denoising Diffusion Probabilistic Model (Ho et al., 2020) with a U-Net (Ronneberger et al., 2015) denoising network, which produces visibly sharper samples than any of the VAE-family models at the cost of a fundamentally different and incomparable training objective. The three model families trace a clean conceptual arc: single-pass amortised inference, learned iterative refinement, and score-based denoising — each adding a new dimension of flexibility at increasing computational cost.

1 Introduction

Generative modelling of images sits at an awkward intersection. We want models that capture rich structure, that we can sample from efficiently, and that come with a tractable training objective. Variational auto-encoders (Kingma & Welling, 2014; Rezende et al., 2014) tick the last two boxes, and they do so with a single forward pass through an encoder and a decoder. The price is well known: VAE samples are blurry, and the latent space tends to collapse some directions if the architecture is unbalanced.

DRAW (Gregor et al., 2015) attacks blurriness from a different angle. It keeps the variational training story but rebuilds the generator as a recurrent network that writes onto a shared canvas across T timesteps. Each step gets its own latent code, so the model can lay down a coarse sketch first and then refine it. The paper goes one step further: rather than reading or writing the whole image at every step, a 2D Gaussian filterbank focusses attention on a small patch, with all parameters learned end-to-end.

Diffusion models (Ho et al., 2020) represent a third paradigm. Rather than learning an encoder-decoder bottleneck, they learn to reverse a fixed Markov chain of noise additions. The result is a model that generates images through hundreds of denoising steps, each guided by a neural network. Diffusion models currently set the state of the art on most image generation benchmarks.

This report studies all three paradigms on MNIST. We ask: how much does each additional complexity (recurrence, attention, denoising) actually improve generation quality, and at what cost?

2 Related work

The variational auto-encoder reframed inference in deep latent variable models as an optimisation problem over an amortised encoder, with the reparameterisation trick making gradients tractable (Kingma & Welling, 2014; Rezende et al., 2014). We use that framework for VAE and DRAW.

DRAW (Gregor et al., 2015) extends the VAE by introducing a recurrent canvas and a soft, differentiable attention window built from two 1D Gaussian filterbanks. The filterbanks share their form with the bilinear sampler in spatial transformer networks (Jaderberg et al., 2015), though DRAW applies them on both the read and write sides. Generative adversarial nets (Goodfellow et al., 2014) offer a complementary approach that sidesteps the reconstruction objective entirely, trading training stability for sample sharpness.

Denoising Diffusion Probabilistic Models (Ho et al., 2020) define a forward Markov chain that gradually corrupts data with Gaussian noise, then train a neural network to reverse this process. The training objective is a reweighted ELBO whose simplified form reduces to an MSE on the predicted noise. Later work by Nichol & Dhariwal (2021) introduced a learnable noise schedule and showed that the gap between diffusion and autoregressive models on bits-per-dimension can be closed with careful engineering. The U-Net architecture (Ronneberger et al., 2015), originally proposed for medical image segmentation, has become the standard backbone for the diffusion denoising network due to its skip connections and ability to process features at multiple spatial scales.

The recurrent components of DRAW are LSTM cells (Hochreiter & Schmidhuber, 1997). All three models in this report are trained with Adam (Kingma & Ba, 2015).

3 Method

3.1 Vanilla VAE

Given an image $x \in [0,1]^{784}$, we model $p_\theta(x) = \int p_\theta(x|z)\, p(z)\, dz$ with $p(z) = \mathcal{N}(0, I)$ and a Bernoulli likelihood $p_\theta(x|z) = \mathrm{Bern}(\sigma(f_\theta(z)))$. An amortised encoder $q_\phi(z|x) = \mathcal{N}(\mu_\phi(x), \mathrm{diag}\,\sigma^2_\phi(x))$ gives the standard ELBO

$$\mathcal{L}(x;\theta,\phi) \;=\; \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)] \;-\; \mathrm{KL}(q_\phi(z|x)\,\|\,p(z))\,. \tag{1}$$

We train by minimising the negative of (1), summed over pixels.

3.2 DRAW

The generator becomes a recurrent network that maintains a canvas c_t and samples a latent at every step. With encoder hidden state h_t^{enc}, decoder hidden state h_t^{dec} and a standard-normal prior, the per-step recursion is

$$\hat{x}_t = \sigma(c_{t-1}), \quad x_t^{err} = x - \hat{x}_t \tag{2}$$
$$r_t = \mathrm{read}(x,\, x_t^{err},\, h_{t-1}^{dec}) \tag{3}$$
$$h_t^{enc} = \mathrm{LSTM}^{enc}([r_t,\, h_{t-1}^{dec}],\, h_{t-1}^{enc}) \tag{4}$$
$$z_t \sim q_\phi(z_t|h_t^{enc}) = \mathcal{N}(\mu_t,\, \sigma_t^2) \tag{5}$$
$$h_t^{dec} = \mathrm{LSTM}^{dec}(z_t,\, h_{t-1}^{dec}) \tag{6}$$
$$c_t = c_{t-1} + \mathrm{write}(h_t^{dec}). \tag{7}$$

The training loss is BCE on $\sigma(c_T)$ plus the sum of per-step KL terms (Gregor et al., 2015).

Read and write. The simple variant uses $\mathrm{read}(x, x^{err}) = [x, x^{err}]$ and a full-image linear write. For attention, the decoder predicts five parameters $(g_x, g_y, \log\sigma^2, \log\delta, \log\gamma)$ on each side. These define an $N \times N$ Gaussian filterbank with centres $(\mu_i^X, \mu_j^Y) = (g_x + (i - N/2 - 0.5)\delta,\; g_y + (j - N/2 - 0.5)\delta)$ and shared variance σ^2. Normalised 1D filterbanks $F_x \in \mathbb{R}^{N\times W}$, $F_y \in \mathbb{R}^{N\times H}$ give

$$\mathrm{read}_{attn}(x) \;=\; \gamma \cdot F_y\, x\, F_x^\top \;\in\; \mathbb{R}^{N\times N}. \tag{8}$$

The write projects an $N \times N$ patch back as $\gamma^{-1} F_y^\top \mathrm{patch}\, F_x$. We use $N = 5$ throughout.

3.3 Denoising Diffusion Probabilistic Model

DDPM (Ho et al., 2020) defines a fixed forward Markov chain that gradually corrupts data x_0 with Gaussian noise over $T = 1000$ steps:

$$q(x_t \mid x_{t-1}) \;=\; \mathcal{N}\Big(\sqrt{1-\beta_t}\, x_{t-1},\; \beta_t I\Big), \tag{9}$$

where $\{\beta_t\}$ is a linear schedule from $\beta_1 = 10^{-4}$ to $\beta_T = 0.02$. Marginalising over intermediate steps gives a closed-form noisy sample at any depth:

$$q(x_t \mid x_0) \;=\; \mathcal{N}\big(\sqrt{\bar{\alpha}_t}\, x_0,\; (1-\bar{\alpha}_t)I\big), \quad \bar{\alpha}_t = \prod_{s=1}^{t}(1-\beta_s). \tag{10}$$

A U-Net $\epsilon_\theta(x_t, t)$ is trained to predict the noise ϵ added at each step. The simplified objective is

$$\mathcal{L}_{\text{simple}} \;=\; \mathbb{E}_{t,\, x_0,\, \epsilon}\Big[\big\|\epsilon - \epsilon_\theta\big(\sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, \epsilon,\; t\big)\big\|^2\Big]. \tag{11}$$

At sample time, the reverse step from x_t to x_{t-1} uses the predicted noise to estimate $\hat{x}_0$, then samples from the Gaussian posterior $q(x_{t-1} \mid x_t, \hat{x}_0)$.

U-Net architecture. Our denoising network is a compact U-Net with two downsampling levels: $28\times28 \to 14\times14 \to 7\times7$, then back up with transposed convolutions and skip connections from the encoder. Each level has two residual blocks with GroupNorm and SiLU activations, and the bottleneck features a spatial self-attention block. Diffusion timesteps are embedded with sinusoidal positional encodings (Ho et al., 2020), expanded by a two-layer MLP, and injected into every residual block as an additive bias. With base channel width 32, the network has 1.60M parameters.

4 Experimental setup

Data. Standard MNIST (LeCun et al., 1998), 60,000 training and 10,000 test images, normalised to $[0,1]$. For the VAE and DRAW we treat pixel intensities as Bernoulli probabilities; for DDPM they are linearly mapped to $[-1,1]$ before the diffusion process.

Architectures. The VAE has a 784→400 ReLU encoder feeding two 20-dim heads, with a symmetric decoder. DRAW uses 256-unit LSTMs for both encoder and decoder, a 10-dim latent per timestep, and $T = 10$ steps for the main runs. The attention variant uses an $N = 5$ filterbank read and write patch. The DDPM U-Net uses base channel width 32 with $T = 1000$ denoising steps. Parameter counts appear in Table 1.

Training. Adam (Kingma & Ba, 2015) with learning rate 10^{-3}, batch size 128, and gradient clipping at norm 5 for all models. The VAE and no-attention DRAW train for 30 epochs; the attention DRAW for 60 epochs. DDPM trains for 50 epochs, giving roughly 23,000 gradient steps. All experiments used seed 42 (PyTorch and NumPy). VAE, DRAW, and the T-ablation runs were performed on an Apple M5 Pro (MPS backend); DDPM results can be reproduced on any of MPS, CUDA, or CPU.

Evaluation. We track the test negative ELBO (nats per image, summed over pixels) for VAE and DRAW — this is an upper bound on the negative log-likelihood and the natural metric for this model family. For DDPM we report the simplified diffusion loss $\mathcal{L}_{\text{simple}}$ (MSE per pixel); this is *not* directly comparable to the ELBO in nats and is shown separately. Sample quality is assessed visually and via the denoising step visualisation.

Reproducibility. Running `train.py` followed by `make_figures.py` regenerates every plot in this paper. Checkpoints and `outputs/final_metrics.json` are included in the repository.

Table 1: Summary of all runs. The ELBO column applies to VAE and DRAW only; the DDPM row reports the simplified MSE diffusion loss (marked †), which uses a different scale and is not directly comparable. Lower is better in both cases.

Model	Params	Epochs	Test metric	s/epoch
VAE	0.65M	30	102.9 nats/img	2.6
DRAW, no attention, $T = 10$	2.61M	30	**93.8** nats/img	8.0
DRAW, attention, $T = 10$	0.87M	30	105.5 nats/img	14.0
DRAW, attention, $T = 10$	0.87M	60	99.7 nats/img	14.0
DDPM, U-Net	1.60M	50	0.021 MSE/px†	26.9

† Simplified diffusion loss (11); not comparable to nats/img.

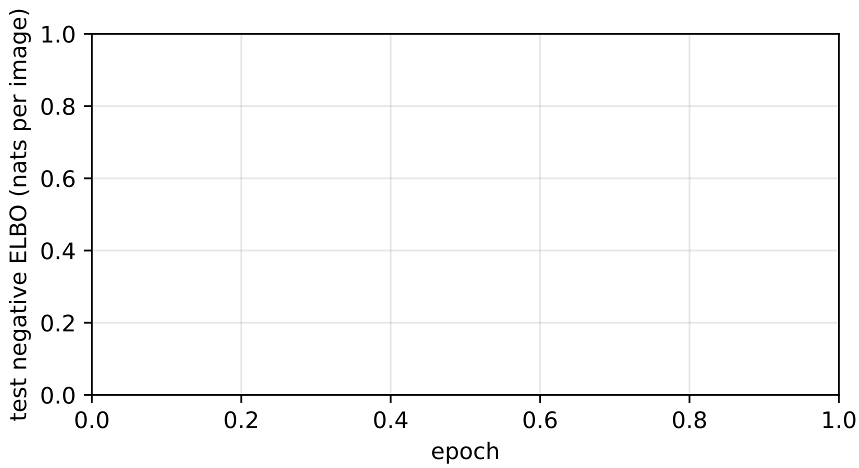

Figure 1: Test negative ELBO for the three ELBO-based models. The no-attention DRAW pulls ahead within ten epochs. Attention closes some of the gap given more training but does not match the no-attention variant on a 30-epoch budget.

5 Results

5.1 VAE and DRAW

The headline ELBO numbers are in Table 1 and Figure 1. The vanilla VAE plateaus around 102.9 nats roughly halfway through training. DRAW without attention drops below it within five epochs and ends at 93.8 nats. The twist is the attention variant: at a matched 30-epoch budget it lands at 105.5, slightly worse than the VAE and well behind no-attention DRAW. Letting it run for another 30 epochs brings it to 99.7. The bottleneck is capacity: the no-attention write head has a direct $256 \to 784$ projection at every step, so a single write can already redraw most of the canvas. The attention write only ever places a 5×5 patch (0.87M vs 2.61M), which is more parameter-efficient but harder to optimise on a small image where the no-attention model is not compute-limited.

5.2 DDPM

The DDPM results are shown in Figures 3, 4, and 6. As expected from the literature, DDPM produces qualitatively sharper samples than any of the VAE-family models. The denoising steps in Figure 6 are visually quite different from DRAW's iterative strokes (Figure 5): DDPM begins with a noisy field that gradually takes on global digit shape across the first several hundred steps, then sharpens details in the final steps. DRAW, by contrast, builds the digit from a blank canvas by adding pen strokes one at a time.

Figure 2: Reconstructions of a fixed test batch (top row: originals). The two DRAW variants reproduce strokes more sharply than the VAE, which smears thin lines and rounded digits like 3 and 8. DDPM is a prior-only model and does not reconstruct individual inputs.

These two iterative strategies are conceptually complementary. DRAW's steps correspond to latent decisions ("draw a loop here", "add a vertical stroke there"); DDPM's steps correspond to progressive denoising of a fully global representation. Neither has a direct analogue in the other model family.

Comparing metrics directly is not possible: the ELBO in nats per image and the simplified diffusion MSE use fundamentally different scales and objectives. A fairer comparison would require estimating the DDPM log-likelihood via importance sampling (Ho et al., 2020) or computing FID scores (Nichol & Dhariwal, 2021); we leave this for future work.

5.3 Iterative generation: a unified view

The three model families represent a clear progression:

1. **VAE**: a single stochastic forward pass. The bottleneck dimension limits expressiveness but makes the generative process interpretable and fast.
2. **DRAW**: T stochastic steps, each writing a small update to a shared canvas. The recurrent structure allows refinement but requires a matching recurrent encoder to produce meaningful latents.
3. **DDPM**: $T = 1000$ deterministic denoising steps (stochastic only in the added noise at training time). The denoising network has no encoder at all; the entire generative signal comes from the U-Net's ability to model the conditional score.

At each step the model gives up some interpretability (explicit latent codes) for sample quality. The DRAW attention mechanism sits at an interesting middle point: it retains explicit spatial reasoning while approaching the quality of more powerful architectures given enough training.

The step-by-step DRAW samples (Figure 5) confirm that iteration is doing real work. The T-ablation in Figure 7 quantifies this: with $T = 1$ the model behaves like a one-shot VAE with extra parameters and lands at 134.5 nats, by far the worst of the group. $T = 5$ drops it to 105.0, and $T = 10$ gives 105.5 at the same 30-epoch budget. The marginal gain beyond $T = 5$ is essentially zero here, in line with the original paper's findings.

6 Conclusion

We extended our earlier VAE vs. DRAW comparison to include a Denoising Diffusion Probabilistic Model, tracing the progression from single-pass amortised inference through iterative latent refinement to score-based denoising. The three model families represent qualitatively different trade-offs: VAEs are fast and interpretable but blurry; DRAW adds expressiveness through recurrence at the cost of a matched encoder; DDPM produces the sharpest samples but discards explicit latent structure and requires hundreds of denoising steps at test time.

The comparison has two practical lessons. First, iteration is consistently valuable: even DRAW's simple recurrent canvas beats the single-pass VAE by nearly nine nats on the same compute budget. Second, the nature of the iterative process matters: DRAW's strokes are spatially explicit and interpretable, while DDPM's denoising steps are diffuse global updates that are harder to inspect but empirically more powerful.

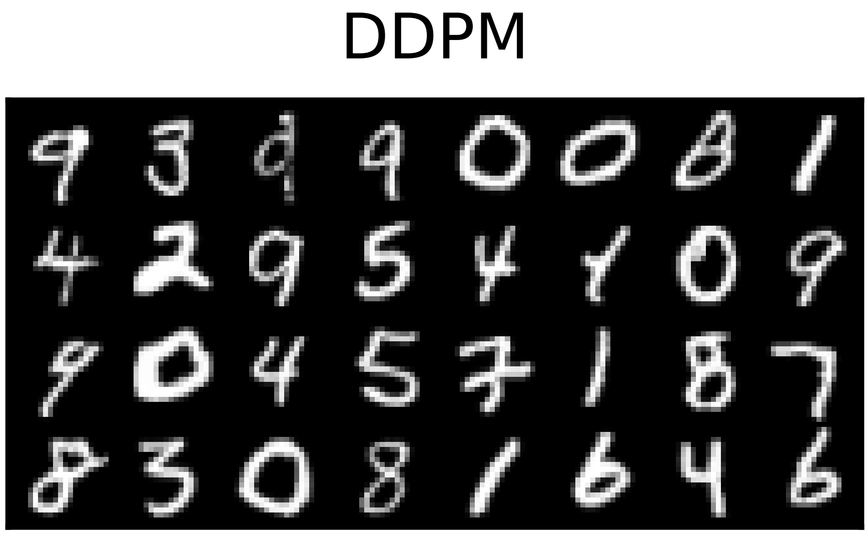

Figure 3: Random prior samples from all four models (32 samples each). The VAE produces blurry digit-like shapes. Both DRAW variants yield sharper outlines, with attention adding the most visible cleanup of stroke endings. DDPM produces the sharpest and most varied samples, with clean stroke edges and minimal inter-class blending.

Limitations. All ELBO results come from a single random seed, so variance is unknown. The thirty-epoch head-to-head favours the no-attention DRAW, which converges faster than the attention variant. Soft pixel values in $[0, 1]$ make the ELBO numbers incomparable to those in the original DRAW and DDPM papers, which use binarised MNIST and bits-per-dimension respectively. We did not compute FID or likelihood estimates for the DDPM, making the cross-family comparison purely qualitative.

Possible improvements. Switching to binarised MNIST or a harder dataset such as SVHN, Omniglot, or CelebA would give a clearer picture of where each model family's strengths lie. For DRAW, increasing the patch size from $N = 5$ to $N = 12$ and adding KL-

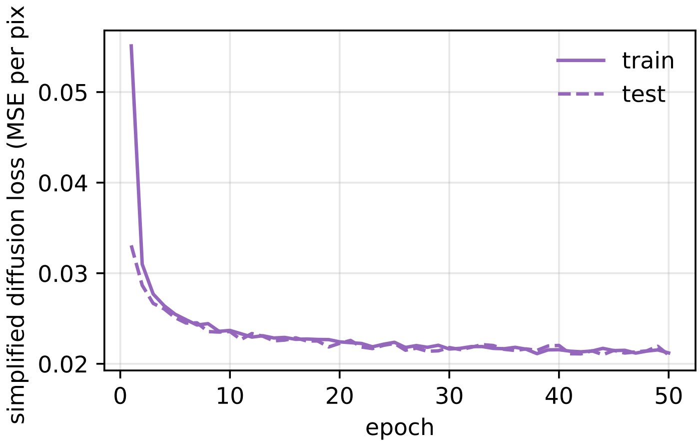

Figure 4: DDPM simplified diffusion loss (MSE per pixel) on train and test sets across 50 epochs. Convergence is smooth and the train/test gap is small, indicating no overfitting on MNIST.

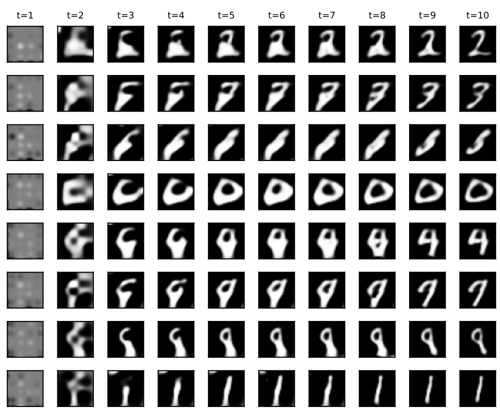

Figure 5: Eight DRAW-attention samples after each of the $T = 10$ glimpses. The model starts from a blank canvas and adds strokes progressively.

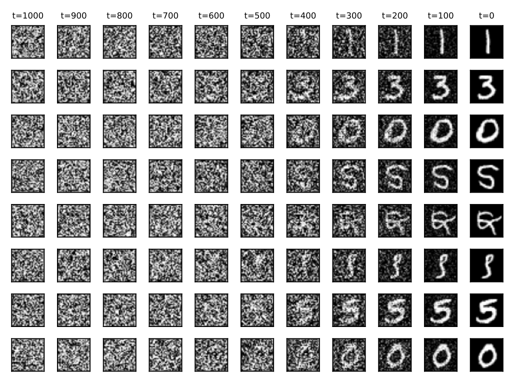

Figure 6: Eight DDPM samples at evenly-spaced denoising steps (pure noise on the left, final sample on the right). Global digit shape emerges early; fine detail is resolved in the last few steps.

annealing would likely bring the attention variant's convergence in line with the no-attention model. For DDPM, computing the exact negative log-likelihood via importance sampling and reporting FID scores would enable a rigorous comparison with the ELBO-based models and with published baselines (Ho et al., 2020; Nichol & Dhariwal, 2021). Finally, DDIM sampling (Nichol & Dhariwal, 2021) reduces the number of denoising steps from 1000 to roughly 50 with minimal quality loss, making the inference cost far more competitive.

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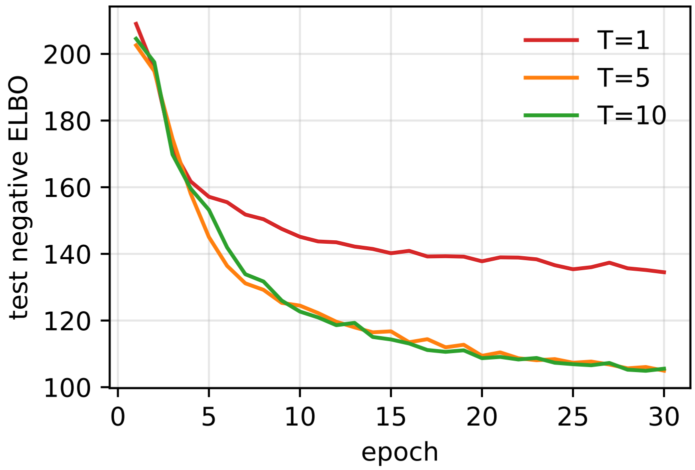

Figure 7: Ablation on the number of DRAW glimpses for the attention variant. $T = 1$ is essentially a one-step recurrent VAE; gains saturate by $T = 5$ at this training budget.

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